
Med-KAG: Medical Knowledge-Augmented Generation

Edouard Haddag, Gabriel H.A. Medeiros, Lina F. Soualmia

LITIS UR 4108, Univ Rouen Normandie, INSA Rouen Normandie

26 May 2026 - EMFI MIE 2026



Motivation: Hallucinations in Medicine

LLMs possess transformative potential for Electronic Health Record (EHR) analysis and diagnosis but suffer from:

- ✓ **Lack of Reliability:** Risk of generating factually incorrect clinical data.
- ✓ **Black-Box Reasoning:** Difficult to audit the source of medical claims.
- ✓ **Clinical Integration:** Clinicians require transparent and trustworthy tools.



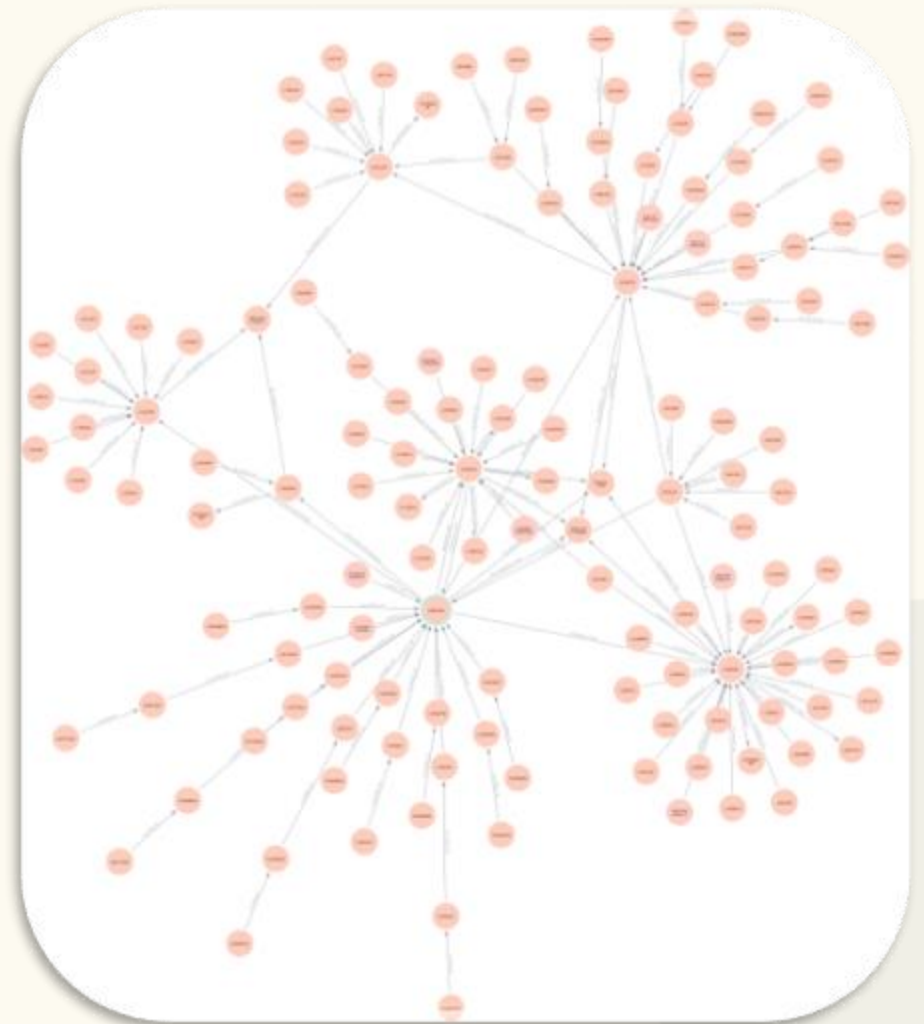
From UMLS to a Knowledge Graph

Medical Foundation

Modeling the UMLS ontology database as a knowledge graph.

- ✓ **Extraction:** Parsing RRF files (MRCONSO, MRREL) from the UMLS Metathesaurus.
- ✓ **Filtering:** Selection of relevant concepts (CUI) and relationships (REL/RELA).
- ✓ **Modeling:** Conversion to a graph-oriented database (Neo4j).

Note: More than 5 million concepts and 30 million interconnected relationships.

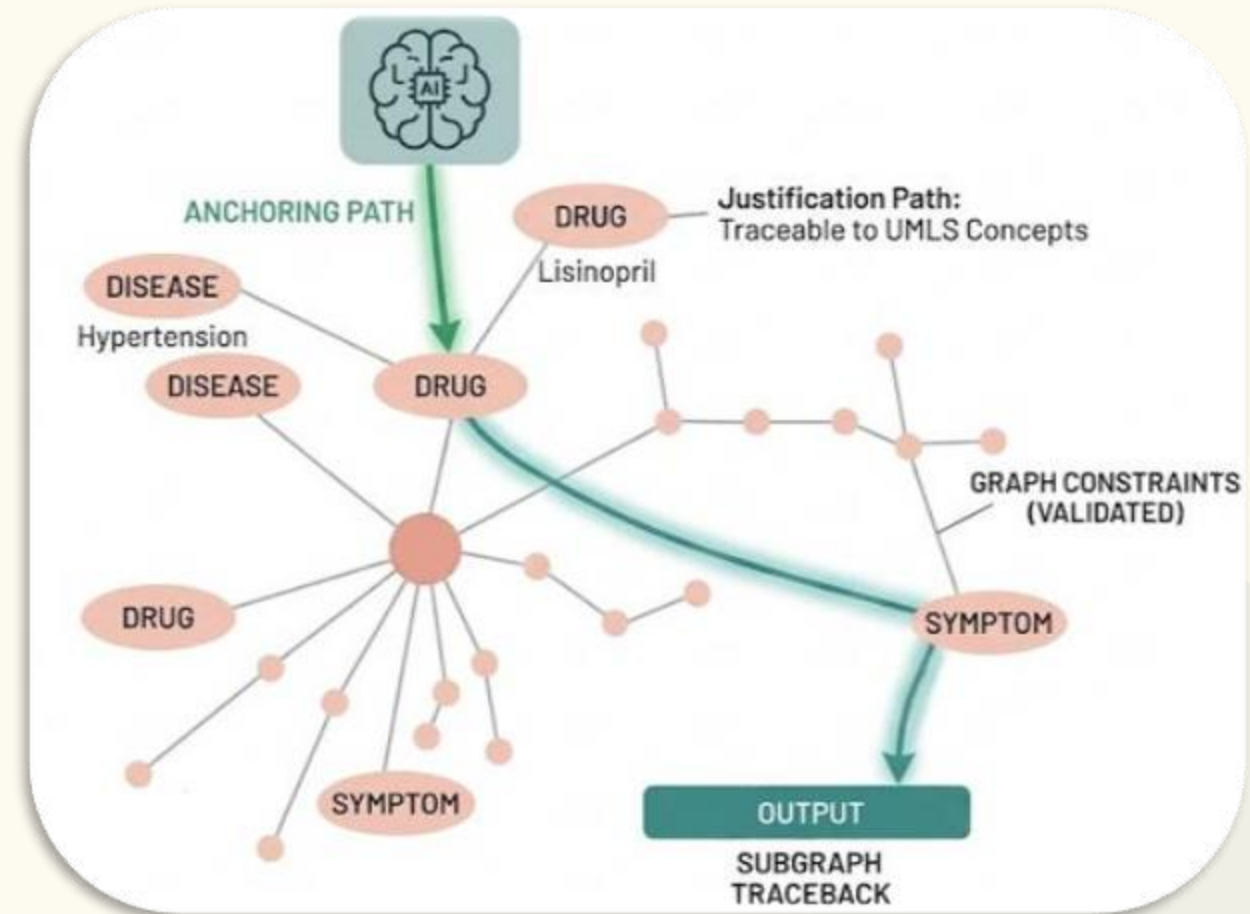


Grounded Medical Retrieval

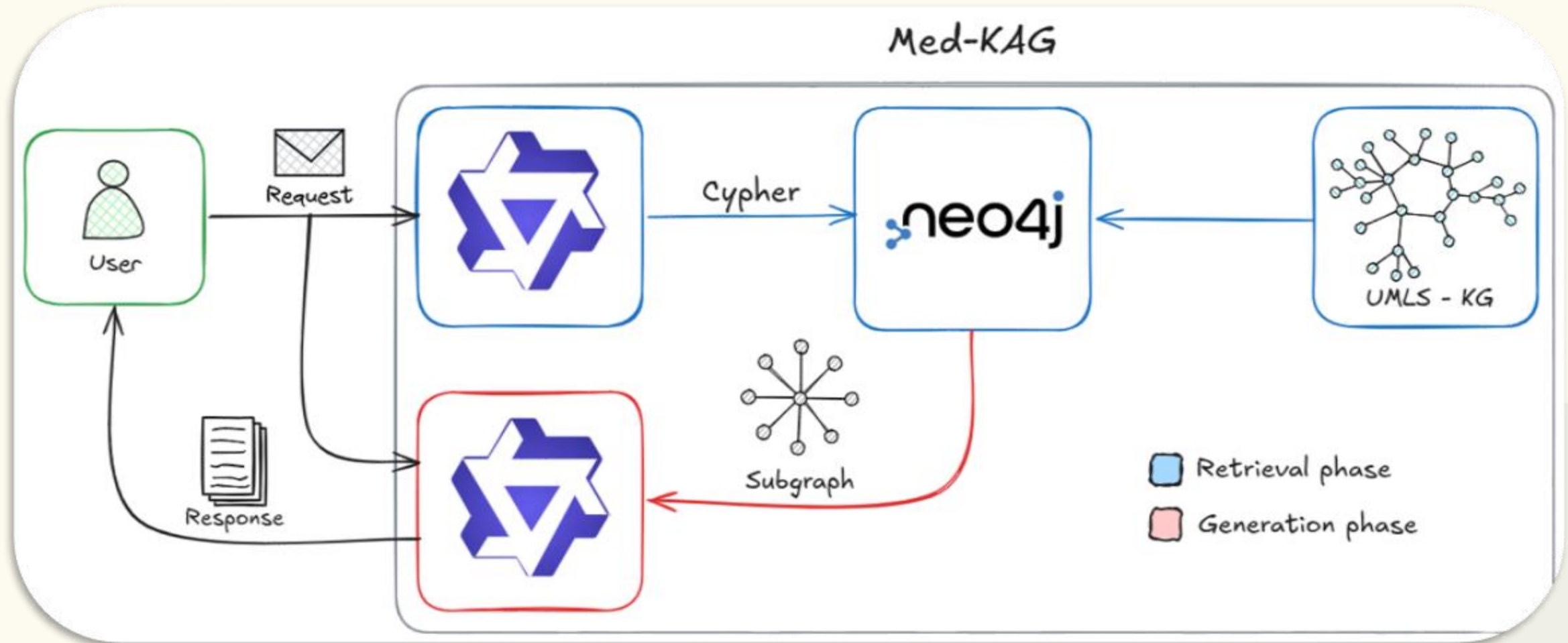
Bridging RAG and KGs

Instead of standard vector databases, we ground generation in structured medical relationships.

- ✓ **UMLS Metathesaurus:** Foundation of domain-specific concepts.
- ✓ **Graph Constraints:** Constrains answers to verified biomedical paths.
- ✓ **Justification:** Traces outputs back to specific UMLS subgraphs.



System Workflow



Preliminary Results on MedQA-US

Question: A junior orthopaedic surgery resident is completing a carpal tunnel repair with the department chairman as the attending physician. During the case, the resident inadvertently cuts a flexor tendon. The tendon is repaired without complication. The attending tells the resident that the patient will do fine, and there is no need to report this minor complication that will not harm the patient, as he does not want to make the patient worry unnecessarily. He tells the resident to leave this complication out of the operative report. Which of the following is the correct next action for the resident to take?

Options:

- (A) Disclose the error to the patient and put it in the operative report,
- (B) Tell the attending that he cannot fail to disclose this mistake,
- (C) Report the physician to the ethics committee,
- (D) Refuse to dictate the operative report

Note: MedQA-US contains 1,273 questions, all of which have four possible answers.

Accuracy on MedQA-US

Configuration	Correct Answers	Accuracy Rate
Native (Qwen3-235B-A22B)	1167 / 1273	91.67%
Med-KAG (RAG + KG)	1162 / 1273	91.28%

Note: Comparable performance despite significant retrieval bottlenecks in this preliminary version.

Retriever Performance Metrics

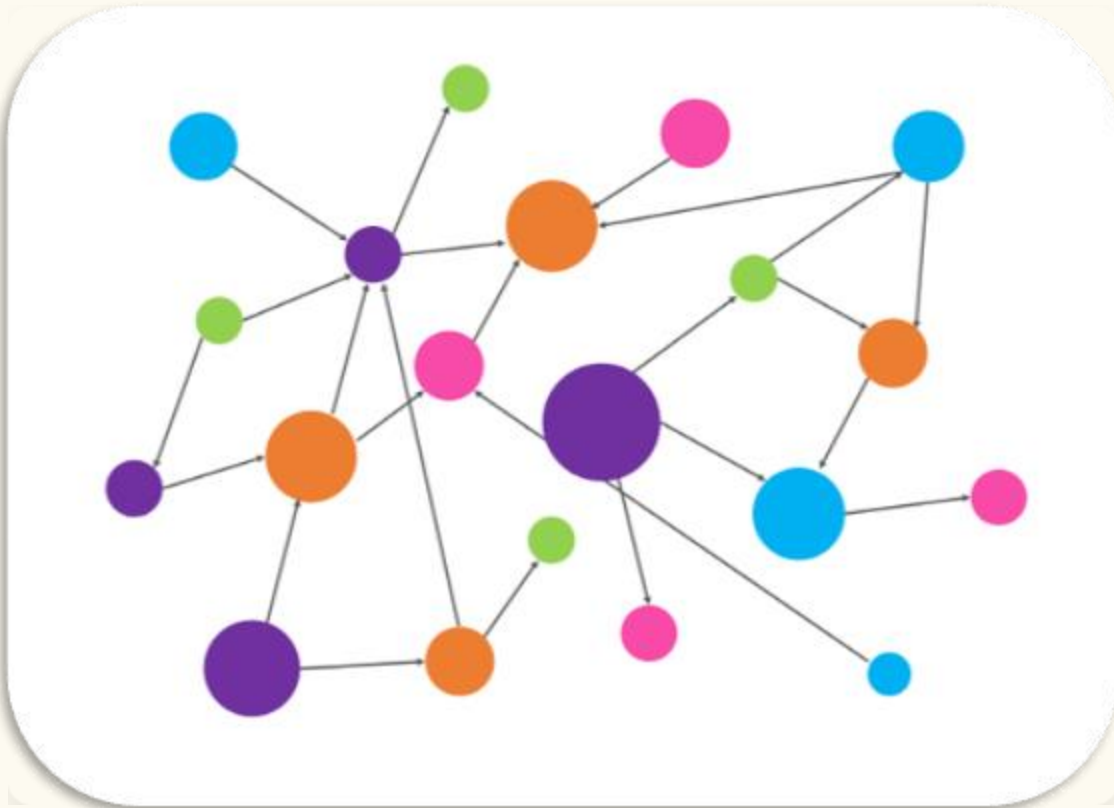
Metric	Mean	Median	Non-Empty
Invalid Cypher query	0.22	0.00	
Retrieved nodes	23.92	0.00	502/1273 (39.4%)
Retrieved edges	27.92	0.00	254/1273 (20.0%)

```
MATCH (c1:CUI)-[r:INVALID_RELATION]-(c2:CUI)
WHERE c1.name CONTAINS "Aspirin"
RETURN c1, r, c2
LIMIT 250
```

```
MATCH (c1 {name CONTAINS "Aspirin"})-[]-(c2)
RETURN c1, c2
LIMIT 250
```

The bottleneck: LLMs struggle with **exact-string matching** for entity linking in Cypher queries, leading to many empty graphs (Median = 0).

Future: Hybrid Retrieval Strategy



Unlocking Potential

- ✓ **Step 1: Embedding-Based Search** to identify "seed" nodes through semantic similarity (solving the exact-match issue).
- ✓ **Step 2: Cypher Traversal** to explore multi-hop relationships starting from seed nodes.
- ✓ **Modular RAG:** Evolution toward self-correction and iterative retrieval cycles.

Thank you for your attention

Edouard Haddag

mie2026@c2fc2f.com

<https://github.com/c2fc2f/Med-KAG/tree/0.1.0>